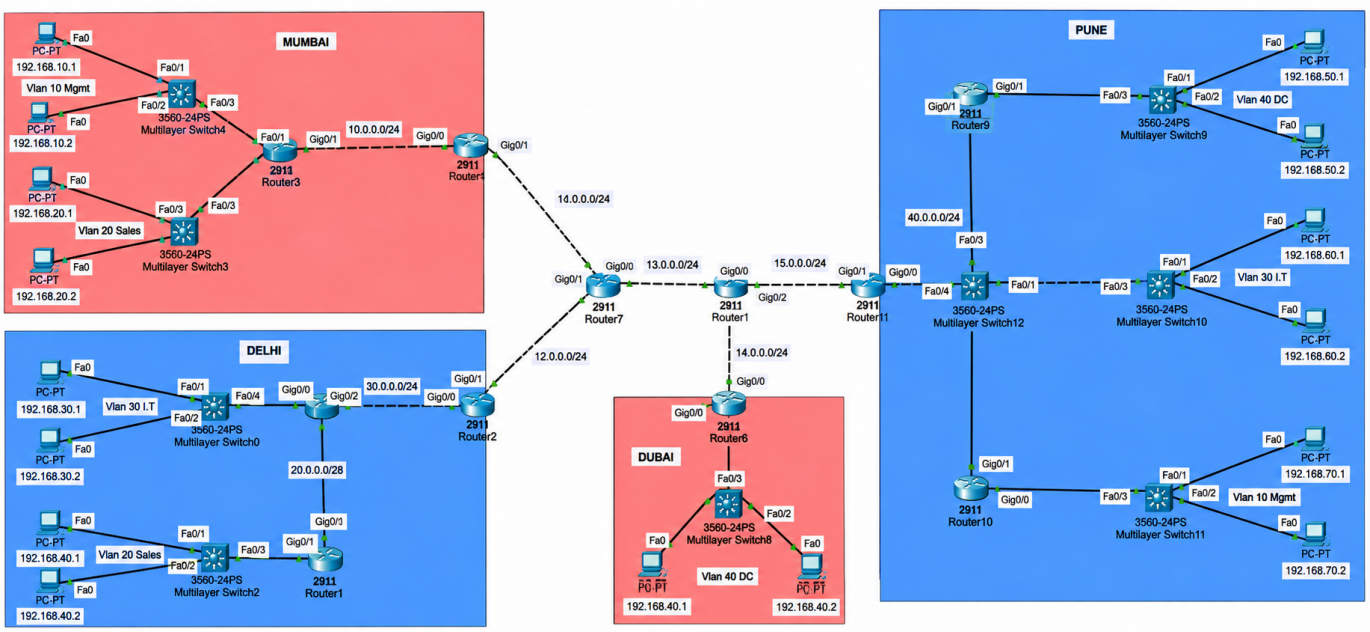
Enterprise Network – Static Routing Documentation

Multi-Site Cisco Network | Mumbai • Pune • Delhi • Dubai



Objective: Build a multi-site enterprise network in Cisco Packet Tracer using static routing so that hosts in Mumbai, Pune, Delhi, and Dubai can communicate across all routed links and VLAN-based LANs.

1. Network Overview

Site	LAN / VLAN Networks	Role
Mumbai	192.168.10.0/24 – VLAN 10 (Mgmt) 192.168.20.0/24 – VLAN 20 (Sales)	Multilayer switches + Router
Pune	192.168.50.0/24 – VLAN 40 (DC) 192.168.60.0/24 – VLAN 30 (IT) 192.168.70.0/24 – VLAN 10 (Mgmt)	Multilayer switches + Router
Delhi	192.168.30.0/24 – VLAN 30 (IT) 192.168.40.0/24 – VLAN 20 (Sales)	Multilayer switches + Routers
Dubai	192.168.40.0/24 – VLAN 40 (DC)	Multilayer switch + Router

2. WAN / Routed Links

Link	Network	Purpose
Mumbai Router ↔ Core	10.0.0.0/24	Mumbai to central routed network
Core ↔ Delhi-side Router	12.0.0.0/24	Core to Delhi path
Core ↔ Central Router	13.0.0.0/24	Core transit link
Central Router ↔ Pune Router	15.0.0.0/24	Transit toward Pune
Central Router ↔ Dubai Router	14.0.0.0/24	Transit toward Dubai
Delhi Router ↔ Delhi Router	30.0.0.0/24	Delhi internal routed connection
Pune Router ↔ Core	14.0.0.0/24 / shown path	Pune transit path
Delhi Router ↔ Core	12.0.0.0/24	Delhi-to-core transit
Pune local router link	40.0.0.0/24	Pune internal routed connection

Note: The topology image is the primary reference for interface names and link addressing. The project uses static routing rather than OSPF, EIGRP, RIP, or another dynamic routing protocol.

3. Configuration Workflow – Small Instructions

• **Step 1 – Build the topology:** Place the routers, multilayer switches, and PCs as shown in the topology. Connect each site to its local LAN and connect the routers using the labeled WAN links.

• **Step 2 – Configure end devices:** Assign each PC an IP address, subnet mask, and the correct default gateway for its VLAN/subnet.

• **Step 3 – Configure VLANs:** Create the required VLANs on the multilayer switches and place the appropriate PC-facing ports into the correct VLANs.

• **Step 4 – Configure Layer-3 gateways:** Configure SVIs on the multilayer switches (where used) and enable Layer-3 routing on the multilayer switches. Give each SVI an address in its local subnet.

• **Step 5 – Configure router interfaces:** Assign IP addresses to every active router interface according to the WAN labels and bring the interfaces up.

• **Step 6 – Verify local connectivity:** From each site, first ping the local default gateway and another PC in the same LAN/VLAN.

• **Step 7 – Add static routes:** On each router and Layer-3 device that must forward remote traffic, add static routes for the destination networks that are not directly connected. Use the appropriate next-hop IP or exit interface.

• **Step 8 – Add return routes:** Static routing requires a working path in both directions. Ensure the remote site also has a route back to the source subnet.

• **Step 9 – Check the routing table:** Use `show ip route` and confirm the remote networks appear with static-route entries marked **S**.

• **Step 10 – Test end-to-end:** Ping PCs across Mumbai, Pune, Delhi, and Dubai. A successful reply confirms that addressing, VLAN gateways, static routes, and return paths are working.

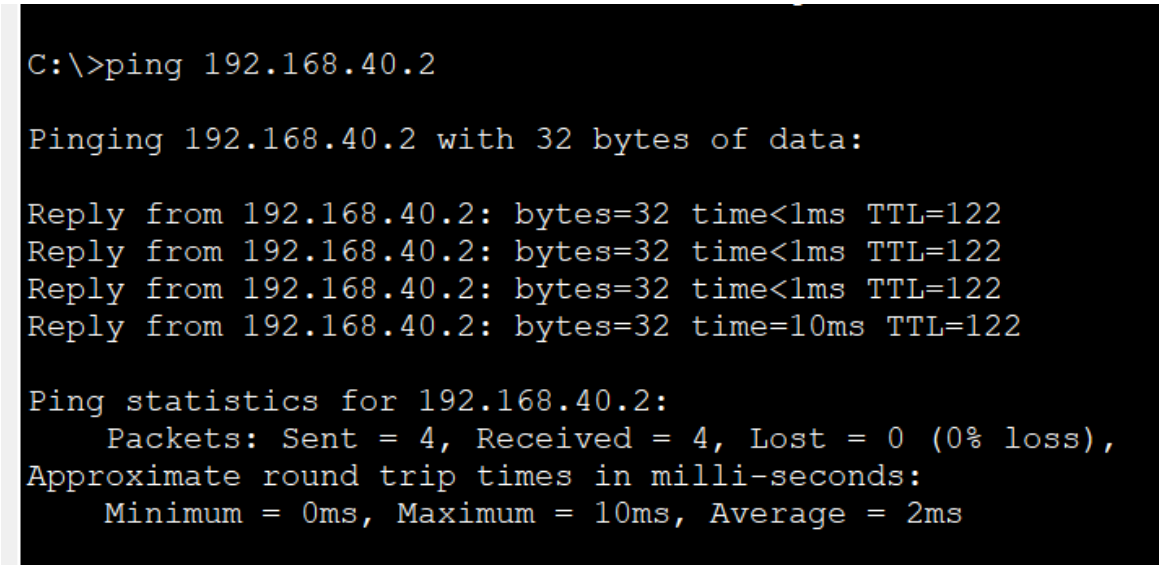
4. Important Static-Routing Commands

Purpose	Command / Example
View interfaces	`show ip interface brief`
View routing table	`show ip route`
Add a static route	`ip route <destination> <mask> <next-hop>`
Add route using exit interface	`ip route <destination> <mask> <exit-interface>`
Test reachability	`ping <destination-ip>`
Trace the path	`traceroute <destination-ip>`
Save configuration	`copy running-config startup-config`

5. Connectivity Verification

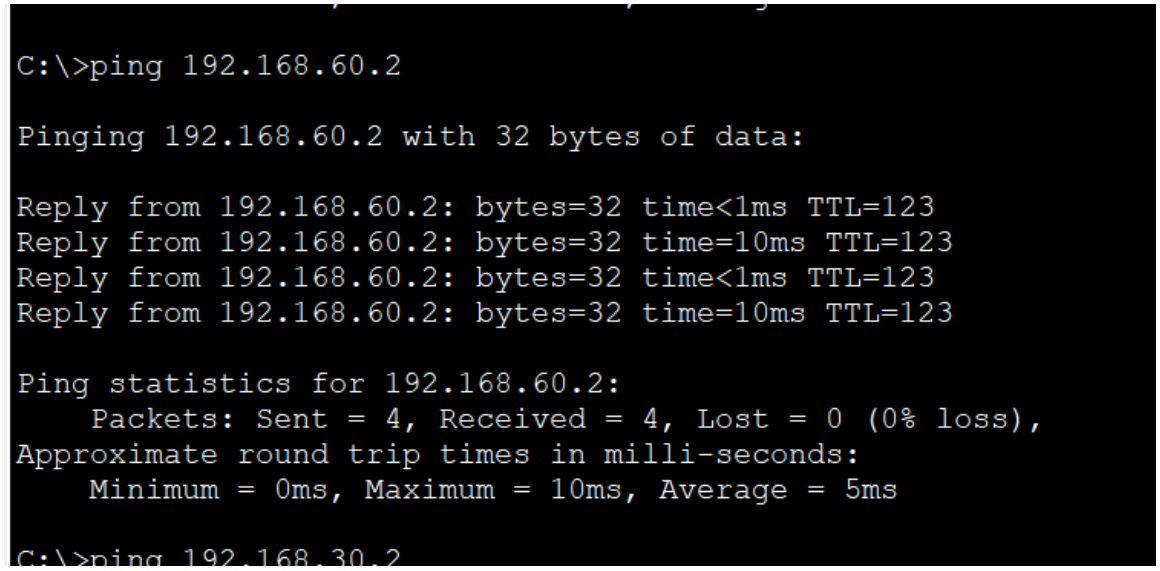
The supplied ping results demonstrate successful end-to-end connectivity between hosts on different site networks. Each test shows 4 packets sent and 4 received with 0% packet loss.

Test 1 – 192.168.40.2



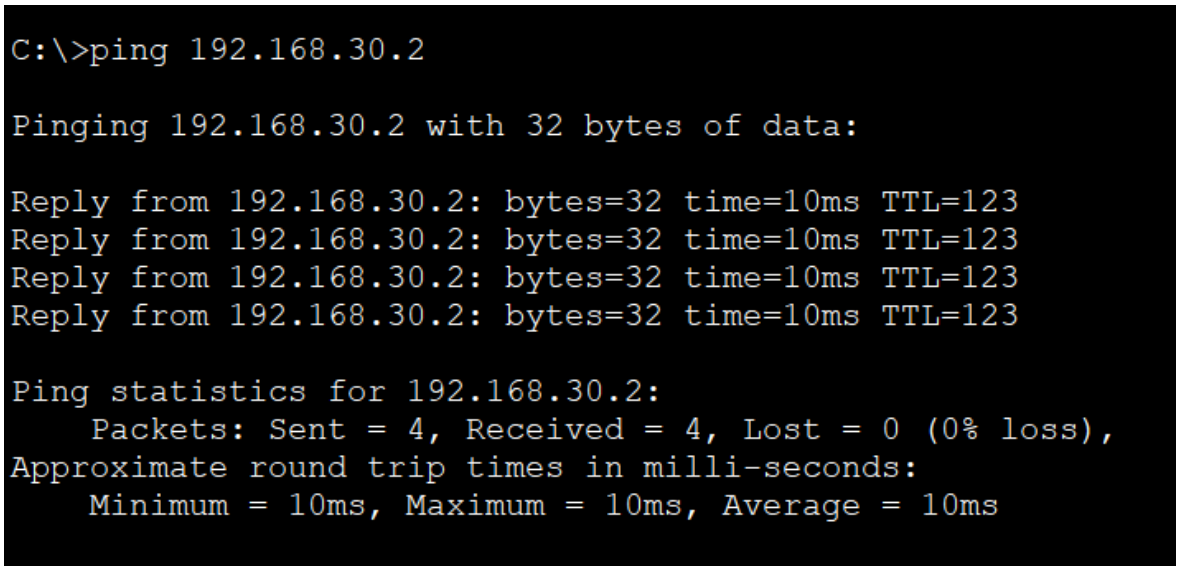
Successful reachability test. Replies were received from 192.168.40.2 with 0% packet loss; average RTT shown was 2 ms.

Test 2 – 192.168.60.2



Successful reachability test. Replies were received from 192.168.60.2 with 0% packet loss; average RTT shown was 5 ms.

Test 3 – 192.168.30.2



Successful reachability test. Replies were received from 192.168.30.2 with 0% packet loss; average RTT shown was 10 ms.

6. Final Verification Checklist

- ■ All router interfaces are configured and operational.
- ■ PC IP addresses and default gateways are correct.
- ■ Required VLANs and Layer-3 gateways are configured.
- ■ Static routes are present for remote networks.
- ■ Return routes are present so replies can travel back.
- ■ `show ip route` confirms static routes with **S** entries.
- ■ Cross-site ping tests complete with 0% packet loss.
- ■ Running configuration is saved to startup configuration.

7. Project Result

Result: The enterprise topology successfully demonstrates inter-site communication using **static routing only**. Hosts located in the Mumbai, Pune, Delhi, and Dubai portions of the topology can reach remote subnets when the required static routes and return routes are configured. The provided ping evidence shows successful connectivity with 0% packet loss for the tested destinations.

Key learning: Static routing is straightforward and predictable for a small or controlled topology, but every remote network and return path must be manually defined. If a link or destination changes, the administrator must update the relevant routes.